EFY TESTED
PREETI BHAKTA

The IoT-Based TIMER SWITCH For Electric Water Heater

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This timer switch for an electric water heater is based on NodeMCU and Blynk IoT cloud solution. An electric water heater is used in bathrooms during the cold season for bathing with hot water. A common problem is that when you wake up in the morning, you need to wait, say 20 minutes, for the electric heater to sufficiently heat the cold water. One possible solution to this problem is using a timer switch.

The timer switch presented here can be used to start and stop the electric water heater from Blynk app in your smartphone. Using this, hot water will be ready when you wake up. The author's prototype is shown in Fig. 1.

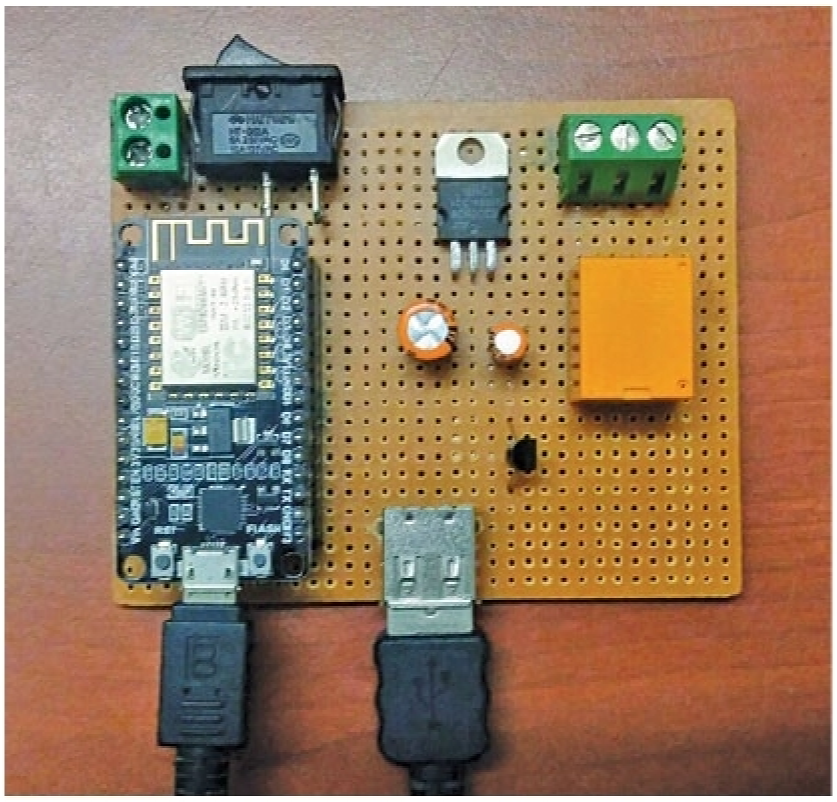

Fig. 1: Author's prototype

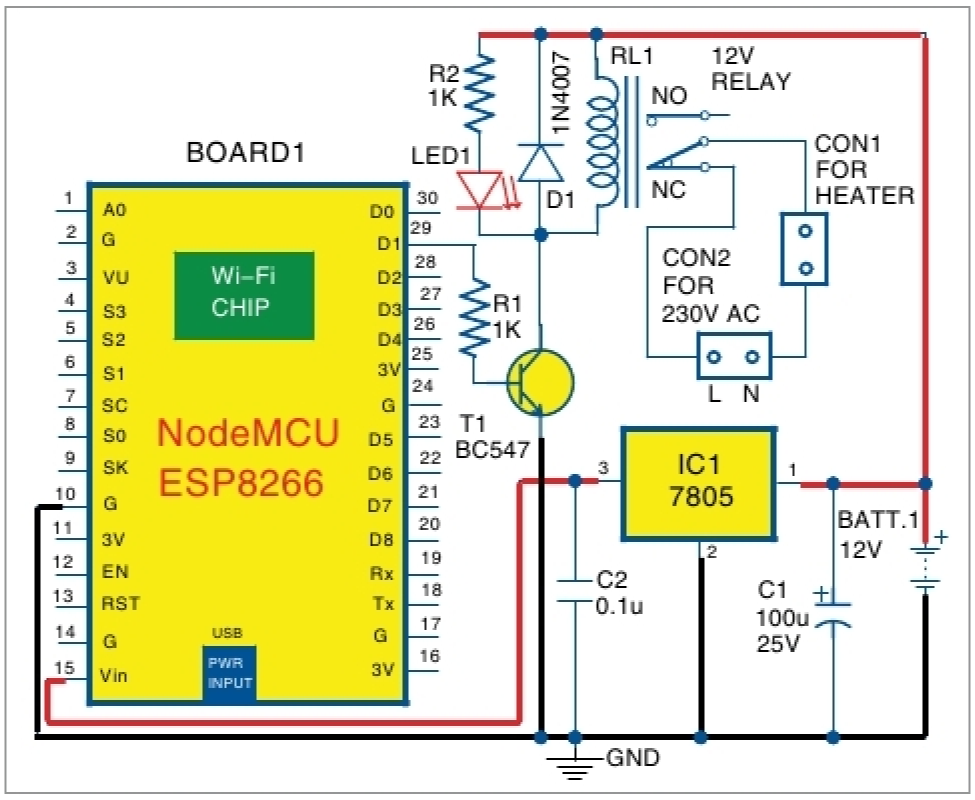

Fig. 2: Circuit diagram of IoT-based timer switch for water heater

Circuit and working

The circuit diagram of the Internet of Things (IoT)-based timer switch for the water heater is shown in Fig. 2. NodeMCU is used as the microcontroller (MCU). The circuit also has 5V voltage regulator 7805 (IC1) for powering NodeMCU. A 12-volt relay (RL1) is used to switch on and switch off the water heater through relay driver transistor BC547 (T1).

NodeMCU is an open source IoT platform that includes firmware that

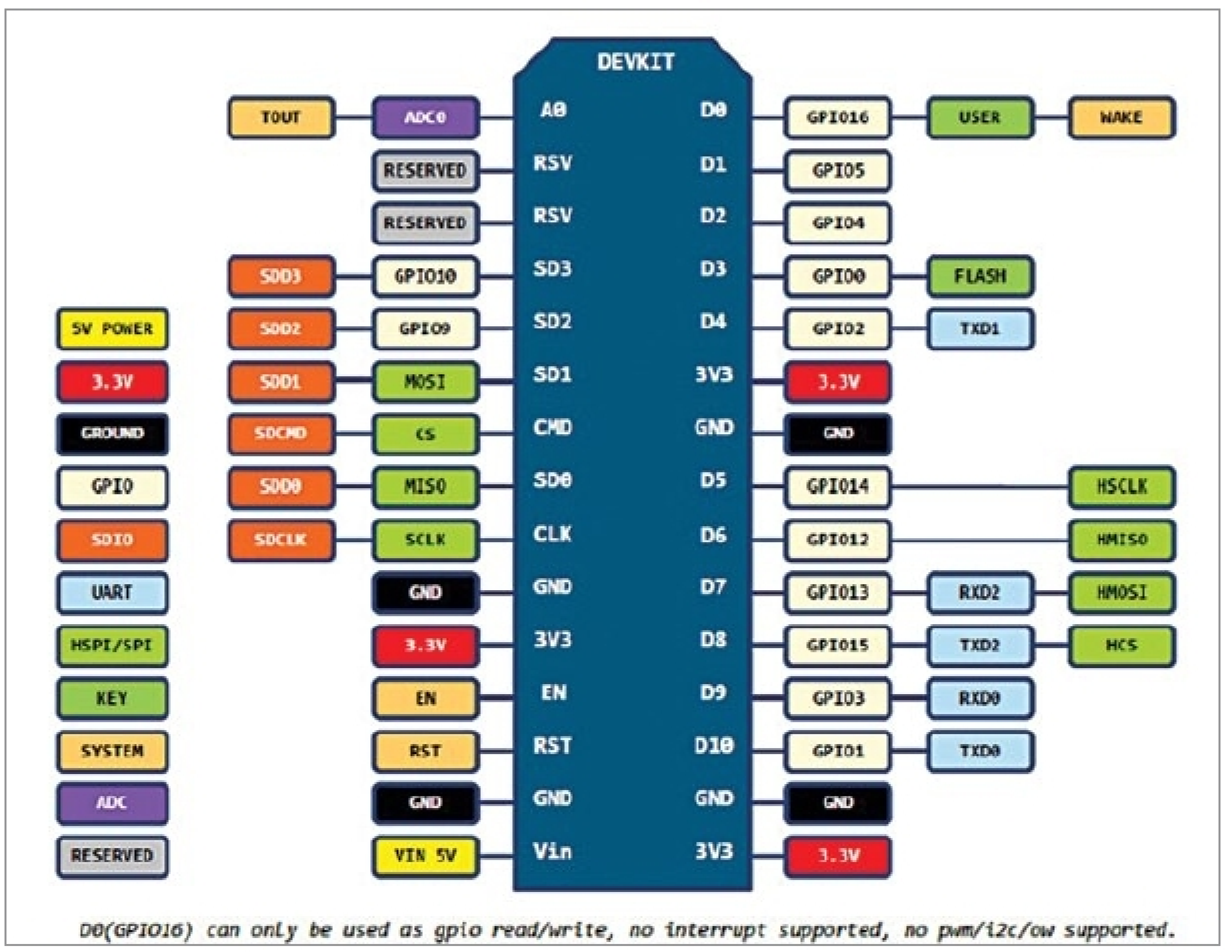

Fig. 3: NodeMCU pin details